Sea-State Aware Ship Detection in Ocean Regions Using Polarimetric Synthetic Aperture Radar

with a Novel Polarimetric Sea-State Index for Adaptive CFAR Kernel Geometry - Group 11

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***Abstract*—Detecting ships reliably in ocean environments is one of the more demanding tasks in remote sensing, with direct relevance to maritime surveillance, navigation safety, and national security. While optical sensors fail in cloudy or nighttime conditions, Synthetic Aperture Radar (SAR) provides consistent, high-resolution imagery under all weather and lighting conditions. Taking this further, Polarimetric SAR (PolSAR) captures the complete electromagnetic scattering matrix, allowing us to go beyond intensity-based analysis and directly reason about *what kind of surface* each pixel corresponds to.**

This paper introduces the Polarimetric Sea-State Index (PSSI), a novel pixel-wise composite indicator built from three physically complementary and mutually non-redundant polarimetric quantities: Cloude–Pottier entropy (H), co-polarization phase difference (ϕ_{HH-VV}), and the Bragg scattering ratio deviation ($\Delta\mathcal{B}$). Together, these three quantities paint a continuous picture of local sea roughness and scattering regime across the entire image. Building directly on PSSI, we propose PSSI-Guided Adaptive CFAR (PGA-CFAR) — a detection framework where both the threshold scaling factor k and the physical *shape and orientation* of the clutter estimation window are dynamically adjusted pixel-by-pixel as a function of the local PSSI gradient. Unlike all existing CFAR variants, which use fixed, isotropic (square or annular) windows, PGA-CFAR deforms its estimation kernel into a rotated ellipse aligned with the estimated local wave-propagation direction, ensuring that background clutter cells are always drawn from statistically homogeneous sea surface regions.

The complete PSSI + PGA-CFAR framework is designed for implementation in MATLAB using full quad-polarimetric SAR datasets. It requires no training data, is grounded in electromagnetic scattering physics, and remains computationally efficient — making it a practical candidate for operational maritime surveillance systems.

***Index Terms*—Polarimetric SAR, Ship Detection, Sea Clutter, CFAR, Polarimetric Sea-State Index (PSSI), Adaptive Kernel Geometry, Bragg Scattering, Cloude–Pottier Decomposition, Maritime Surveillance**

I. Introduction

The ocean covers more than 70% of Earth's surface, and a significant fraction of global trade, resource extraction, and geopolitical activity takes place on it. Monitoring this vast space — detecting illegal fishing, tracking cargo vessels, responding to distress situations, or identifying unauthorized maritime movements — demands imaging systems that work consistently regardless of time of day or atmospheric conditions [1]. Traditional optical satellites, while effective in clear conditions, are rendered largely useless by cloud cover, which is a persistent feature of ocean environments. SAR solves this problem at its root: because microwave radiation penetrates clouds and does not depend on solar illumination, SAR sensors can image the ocean continuously and globally [2].

Even so, detecting ships in SAR imagery is far from straightforward. The central challenge is the behavior of *sea clutter* — the radar backscatter from the ocean surface itself. The ocean is not a static background: it is a dynamic, spatially heterogeneous medium whose radar reflectivity varies with wind speed, wave height, swell direction, foam coverage, and incidence angle [3]. A ship that stands out clearly against a calm sea may be nearly indistinguishable from a rough wave patch in high sea states. Standard intensity-based detection methods, and even classical CFAR detectors, struggle with this variability because they assume the background clutter statistics are uniform within the estimation window — an assumption that real ocean scenes routinely violate.

Polarimetric SAR offers a principled solution. By transmitting and receiving in multiple polarization states simultaneously, PolSAR captures not just the strength of the radar

return but its full electromagnetic character. Ships — built from metal, full of right-angle junctions between hull and deck, hull and sea surface, and superstructure elements — produce scattering that is physically distinct from ocean waves. The dominant ship scattering mechanism is double-bounce (dihedral-type) reflection, which produces characteristic polarimetric signatures: low entropy, large mean alpha angle ($\alpha \to 90°$), and co-polarization phase differences near $180°$. Ocean surfaces, by contrast, are governed at typical microwave incidence angles by Bragg resonance — a surface scattering mechanism with entirely different polarimetric fingerprints [4].

The gap this work addresses is the following: while many researchers have used individual polarimetric features (entropy, or intensity ratio, or phase difference) to improve ship detection, no prior method has (a) combined all three major independent polarimetric ocean scattering indicators into a single, physically-motivated sea-state map, nor (b) used that map to *deform the geometry* of the CFAR estimation kernel rather than just adjusting its threshold. This paper proposes both: the **Polarimetric Sea-State Index (PSSI)** and the **PSSI-Guided Adaptive CFAR (PGA-CFAR)** detector. The result is a system that knows not only *how rough* the sea is at each pixel, but also *in which direction* the sea state is changing — and uses both pieces of information to make smarter detection decisions.

II. Related Work

A. Intensity-Based and CFAR Detectors

Early ship detection in SAR relied almost entirely on intensity thresholding: ships appear brighter than the surrounding ocean in many configurations, so a simple amplitude threshold can isolate them. The obvious problem is that the appropriate threshold depends on local clutter level, which varies across the scene. The CFAR framework, introduced in the radar context by Finn and Johnson [5], addresses this by estimating the local clutter statistics from a sliding reference window and setting the threshold adaptively. Cell-Averaging CFAR (CA-CFAR) [6] is the most widely used variant and forms the baseline that this work extends.

Considerable effort has gone into developing CFAR variants that are more robust to non-Gaussian and heterogeneous clutter. The Weibull and K-distribution models have been fitted to ocean clutter [7]; OS-CFAR [8] handles outliers via ordered statistics; and two-parameter CFAR [9] separately estimates the shape and scale of the clutter distribution. However, a characteristic shared by *all* of these variants is the use of a fixed, isotropic (square or ring-shaped) estimation window. This paper is the first to replace that fixed geometry with a directionally adaptive elliptical kernel driven by a polarimetric sea-state map.

B. Polarimetric Features for Ship Detection

The use of PolSAR for ship detection was systematically explored following the availability of spaceborne quad-pol sensors such as Envisat ASAR, RADARSAT-2, and TerraSAR-X. Touzi et al. [13] demonstrated that the coherency matrix provides significantly richer discrimination between ships and sea clutter than single-channel intensity. Cloude and Pottier's entropy/alpha decomposition [10] was applied to ship detection by several groups, establishing that ships occupy a distinctly different region of the H/α plane than ocean surfaces.

Cameron and Rais [12] explored symmetric scattering component analysis for target characterization; Freeman and Durden [11] introduced the three-component decomposition separating surface, double-bounce, and volume scattering. Specific to maritime applications, Nunziata et al. [14] examined the co-polarization ratio and phase difference as ship discriminators in dual-pol data, confirming the near-$180°$ CPD signature of ship targets. Velotto et al. [15] demonstrated ship detection improvements using TerraSAR-X dual-pol imagery over single-channel data.

More recently, Singha et al. [16] proposed entropy-based adaptive thresholding, where the detection threshold scales with local polarimetric entropy as a proxy for sea roughness. This is the closest precursor to our approach: we extend their single-feature adaptation to a three-feature fusion (PSSI) and additionally adapt the kernel *geometry*, which Singha et al. did not address.

C. Learning-Based Approaches

Deep learning has produced strong ship detection results in both SAR and PolSAR imagery [17], [18]. Convolutional neural network (CNN) detectors trained on large annotated datasets can implicitly learn complex clutter patterns and achieve high detection rates. The limitations are well-known: labelled SAR ship datasets remain scarce, domain shift between sensors is substantial, and the resulting models are opaque to physical interpretation. For operational maritime surveillance systems, where decisions must be auditable and sensor conditions may change, a physics-based approach that does not require training data offers clear practical advantages.

III. Polarimetric SAR Fundamentals

This section establishes the mathematical framework underlying the proposed method. Readers familiar with PolSAR fundamentals may proceed directly to Section IV.

A. Scattering Matrix

When a radar pulse illuminates a target, the relationship between the transmitted and received electric field vectors is described by the complex scattering matrix:

$$\mathbf{S} = \begin{bmatrix} S_{HH} & S_{HV} \\ S_{VH} & S_{VV} \end{bmatrix}, \tag{1}$$

where each element S_{pq} is a complex number representing the amplitude and phase of the return when transmitting in polarization q and receiving in polarization p (H = horizontal, V = vertical). For a monostatic radar observing reciprocal (non-rotating) targets, $S_{HV} = S_{VH}$, which reduces the independent complex measurements from four to three.

B. Target Vectors and Second-Order Statistics

Since the scattering matrix is a random variable over a distributed surface target, we work with its second-order statistics. Two equivalent formulations are standard. The *lexicographic target vector*:

$$\mathbf{k}_L = \begin{bmatrix} S_{HH} & \sqrt{2}S_{HV} & S_{VV} \end{bmatrix}^T \tag{2}$$

generates the *covariance matrix* $\mathbf{C} = \langle \mathbf{k}_L \mathbf{k}_L^\dagger \rangle$. The *Pauli target vector*:

$$\mathbf{k}_P = \frac{1}{\sqrt{2}} \begin{bmatrix} S_{HH} + S_{VV} \\ S_{HH} - S_{VV} \\ 2S_{HV} \end{bmatrix} \tag{3}$$

generates the *coherency matrix*:

$$\mathbf{T} = \langle \mathbf{k}_P \mathbf{k}_P^\dagger \rangle = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix}, \tag{4}$$

where $\langle \cdot \rangle$ denotes spatial ensemble averaging and $(\cdot)^\dagger$ is the Hermitian (conjugate) transpose. Both $\mathbf{C}$ and $\mathbf{T}$ are 3×3 Hermitian positive semi-definite matrices of rank at most 3. $\mathbf{T}$ is preferred in practice because its diagonal elements directly correspond to physically interpretable Pauli scattering powers.

C. Pauli Decomposition

The Pauli basis provides the most intuitive decomposition of the scattering matrix into canonical mechanisms:

$$\mathbf{S} = \frac{a}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{b}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + \frac{c}{\sqrt{2}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \tag{5}$$

where $a = (S_{HH} + S_{VV})/\sqrt{2}$, $b = (S_{HH} - S_{VV})/\sqrt{2}$, and $c = \sqrt{2}S_{HV}$. The powers $|a|^2$, $|b|^2$, and $|c|^2$ represent single-bounce surface scattering, double-bounce (dihedral) scattering, and volume (diffuse) scattering contributions, respectively. Forming a false-color composite image with $\{|b|^2, |c|^2, |a|^2\} \to \{\text{R, G, B}\}$ immediately reveals ship structures as bright red features against a blue-green ocean background — an intuition that motivates using $|b|^2$ as a ship discriminator.

D. Cloude–Pottier Eigendecomposition

The coherency matrix $\mathbf{T}$ admits the spectral decomposition [10]:

$$\mathbf{T} = \sum_{i=1}^{3} \lambda_i \, \mathbf{u}_i \mathbf{u}_i^\dagger, \tag{6}$$

where $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0$ are real eigenvalues and $\mathbf{u}_i$ are orthonormal eigenvectors. Define pseudo-probabilities $p_i = \lambda_i / \sum_j \lambda_j$ with $\sum_i p_i = 1$.

Polarimetric entropy measures the statistical randomness of the scattering process:

$$H = -\sum_{i=1}^{3} p_i \log_3 p_i, \quad H \in [0, 1]. \tag{7}$$

A single dominant scattering mechanism (e.g., a specular reflector) gives $H \approx 0$. A fully random mixture of mechanisms gives $H \approx 1$. In practice, calm ocean surfaces have $H \approx 0.3$–0.5, rough seas have $H \approx 0.6$–0.8, and metallic ship structures have $H \approx 0.1$–0.3.

Anisotropy discriminates between the second and third scattering mechanisms:

$$A = \frac{\lambda_2 - \lambda_3}{\lambda_2 + \lambda_3}, \quad A \in [0, 1]. \tag{8}$$

Mean alpha angle encodes the dominant scattering mechanism type:

$$\alpha = \sum_{i=1}^{3} p_i \alpha_i, \quad \alpha \in [0^\circ, 90^\circ], \tag{9}$$

where α_i is the angle of eigenvector $\mathbf{u}_i$ relative to the surface scattering eigenvector. Bragg ocean scattering sits near $\alpha \approx 0^\circ$–20°; ship double-bounce scattering drives α toward 90°.

E. Co-Polarization Phase Difference

The co-polarization phase difference (CPD) is:

$$\phi_{HH-VV} = \angle \langle S_{HH} S_{VV}^* \rangle, \tag{10}$$

where $(\cdot)^*$ denotes complex conjugation. For a Bragg-scattering ocean surface, the Small Perturbation Model [19] predicts $\phi_{HH-VV} \approx 0^\circ$ at low roughness, with increasing deviation as roughness exceeds the Bragg validity limit. Ship targets, due to their dihedral corner-reflector geometry, produce ϕ_{HH-VV} close to 180°. This near-180° signature makes CPD one of the most powerful available ship–sea discriminators [14].

F. Bragg Scattering and Surface Roughness

At typical SAR microwave frequencies and incidence angles (20°–60°), ocean radar backscatter is dominated by Bragg resonance [20]: the radar couples to ocean surface capillary-gravity waves whose spatial wavelength Λ_B satisfies:

$$\Lambda_B = \frac{\lambda_r}{2 \sin \theta_i}, \tag{11}$$

where λ_r is the radar wavelength and θ_i is the local incidence angle. The Bragg model (Small Perturbation Model) predicts backscatter coefficients:

$$\sigma_{HH}^0 \propto |R_{HH}(\theta_i)|^2 W(\kappa_B), \quad \sigma_{VV}^0 \propto |R_{VV}(\theta_i)|^2 W(\kappa_B), \tag{12}$$

where R_{HH} and R_{VV} are the polarization-dependent Fresnel reflection coefficients of the sea surface, and $W(\kappa_B)$ is the ocean wave spectral density at the Bragg wavenumber $\kappa_B = 2k_r \sin \theta_i$. Under the Bragg model, the ratio of HH to VV backscatter is a deterministic function of incidence angle and sea surface dielectric constant:

$$\mathcal{B}_{\text{pred}}(\theta_i) = \frac{\sigma_{HH}^0}{\sigma_{VV}^0} = \left| \frac{R_{HH}(\theta_i)}{R_{VV}(\theta_i)} \right|^2. \tag{13}$$

When the measured ratio $\mathcal{B}_{\text{meas}}$ deviates significantly from $\mathcal{B}_{\text{pred}}$, it signals that the Bragg model is breaking down — either because of high sea roughness (non-linear wave breaking, whitecapping) or because the pixel contains a non-ocean target such as a ship.

IV. Novel Contribution: Polarimetric Sea-State Index (PSSI)

A. Motivation

The core insight motivating PSSI is that the ocean surface, under varying sea states, leaves distinct and *separable* signatures across multiple polarimetric observables — and that these signatures are physically orthogonal: each captures a different facet of the scattering process that the others cannot recover. Specifically:

- **Entropy** H measures overall scattering *complexity*. A rough sea produces multiple co-existing scattering mechanisms (Bragg waves, breaking waves, foam, specular reflections from facets), driving H upward. However, entropy alone cannot distinguish between a rough sea and a scene contaminated by speckle or mixed land–ocean pixels.
- **CPD** ϕ_{HH-VV} measures the deviation of co-polarization phase from the Bragg prediction. As sea roughness exceeds the Bragg validity limit, non-Bragg contributions (specular facets, tilted surfaces) distort ϕ_{HH-VV} away from $0°$. However, CPD is a phase quantity and therefore noisy for single-look data, and it carries a $\pm\pi$ ambiguity.
- **Bragg ratio deviation** $\Delta\mathcal{B}$ measures the departure of the measured HH/VV power ratio from its theoretically predicted Bragg value. This captures roughness-induced changes in the *magnitude* balance between polarization channels. However, $\mathcal{B}$ is sensitive to incidence angle variation and requires accurate sensor metadata.

No single feature is sufficient; their combination is. The PSSI fuses all three into a single, continuous, per-pixel sea-roughness score.

B. PSSI Definition

We define the **Polarimetric Sea-State Index** at pixel (m, n) as the weighted linear combination:

$$\boxed{\mathrm{PSSI}(m,n) = w_1 \cdot H(m,n) \;+\; w_2 \cdot \Phi(m,n) \;+\; w_3 \cdot \Delta\mathcal{B}(m,n)} \tag{14}$$

with weights $w_1, w_2, w_3 \geq 0$ satisfying $w_1 + w_2 + w_3 = 1$. Each component is normalized to $[0, 1]$:

Component 1 — Entropy term $H(m, n)$**:** Computed directly from the eigendecomposition of $\mathbf{T}$ via (7). Already in $[0, 1]$ by definition.

Component 2 — Normalized CPD deviation $\Phi(m, n)$**:**

$$\Phi(m,n) = \frac{|\phi_{HH-VV}(m,n)|}{\pi} \;\in\; [0,1]. \tag{15}$$

$\Phi \approx 0$ for a Bragg-dominated calm ocean surface; $\Phi \to 1$ as the CPD approaches $\pm 180°$ (ship or very rough sea).

Component 3 — Normalized Bragg ratio deviation $\Delta\mathcal{B}(m, n)$**:**

$$\Delta\mathcal{B}(m,n) = \min\left(1,\; \frac{\left|\mathcal{B}_{\text{meas}}(m,n) - \mathcal{B}_{\text{pred}}(\theta_i)\right|}{\mathcal{B}_{\text{pred}}(\theta_i)}\right), \tag{16}$$

where $\mathcal{B}_{\text{pred}}(\theta_i)$ is evaluated at the local incidence angle from the sensor's incidence angle map. The $\min(\cdot, 1)$ clamp prevents saturation artifacts at very high roughness or near ships.

Default weight assignment: $w_1 = 0.4$, $w_2 = 0.3$, $w_3 = 0.3$, reflecting the higher robustness of entropy relative to the phase and ratio measurements. These may be tuned per sensor configuration in Phase II.

C. Physical Interpretation

The PSSI map provides a continuous, spatially resolved picture of sea-state severity:

- $\mathrm{PSSI} \approx 0$: Calm, Bragg-dominated ocean. Low entropy, near-zero CPD, ratio matches Bragg prediction. Excellent detection conditions — small CFAR threshold sufficient.
- $\mathrm{PSSI} \approx 0.5$: Moderate sea state. Emerging non-Bragg contributions. Detection remains feasible but threshold must be raised.
- $\mathrm{PSSI} \approx 1$: Very rough sea. High randomness, large CPD deviation, strongly non-Bragg. Challenging for any detector — aggressive threshold adaptation required.

A key property: ship targets do *not* produce high PSSI. Although ships exhibit large Φ and $\Delta\mathcal{B}$, their extremely low entropy ($H \ll 0.3$ for coherent dihedral structures) suppresses PSSI below the high-roughness sea level. This means PSSI is a *sea-specific* indicator rather than a generic "anomaly" detector, and does not self-report ship pixels as rough sea.

D. PSSI Gradient: Estimating Wave Direction

To drive the geometry adaptation in PGA-CFAR, we compute the spatial gradient of the PSSI field using a Sobel operator:

$$\nabla\mathrm{PSSI}(m,n) = \left(\frac{\partial\,\mathrm{PSSI}}{\partial m},\; \frac{\partial\,\mathrm{PSSI}}{\partial n}\right). \tag{17}$$

The gradient direction:

$$\theta_\nabla(m,n) = \arctan\left(\frac{\partial\,\mathrm{PSSI}/\partial n}{\partial\,\mathrm{PSSI}/\partial m}\right) \tag{18}$$

estimates the orientation of the dominant sea-state transition, which in practice aligns with the wave propagation direction. The gradient magnitude:

$$\|\nabla\mathrm{PSSI}\|(m,n) = \sqrt{\left(\frac{\partial\,\mathrm{PSSI}}{\partial m}\right)^2 + \left(\frac{\partial\,\mathrm{PSSI}}{\partial n}\right)^2} \tag{19}$$

quantifies how rapidly the sea state is changing. Large gradient magnitudes indicate clutter transition zones — precisely the regions where conventional isotropic CFAR windows straddle two different statistical populations, resulting in poorly estimated thresholds and elevated false alarms.

V. PSSI-Guided Adaptive CFAR (PGA-CFAR)

A. Why Conventional CFAR Fails in Heterogeneous Seas

Standard CA-CFAR computes the detection threshold as:

$$T_{\text{CA}} = \hat{\mu}_{\text{ref}} + k_{\text{fixed}} \cdot \hat{\sigma}_{\text{ref}}, \tag{20}$$

where $\hat{\mu}_{\text{ref}}$ and $\hat{\sigma}_{\text{ref}}$ are estimated from a fixed rectangular or annular background window. Three assumptions underlie this:

(1) the clutter is homogeneous within the window; (2) clutter statistics are isotropic (no preferred spatial direction); and (3) k_{fixed} is an appropriate constant everywhere in the image. In real ocean scenes:

Assumption (1) fails at sea-state boundaries, where the background window straddles regions of different roughness, causing the estimated mean to be inflated and the threshold to miss targets in the calmer region. Assumption (2) fails because ocean waves impose a directional structure on clutter statistics — cells along wave crests share more statistical similarity than cells across wave crests. Assumption (3) fails because the appropriate false-alarm-controlling threshold is higher in rough seas than in calm seas.

PGA-CFAR corrects all three failures simultaneously.

B. Adaptive Threshold Scaling

The first adaptation is to replace the fixed k with a PSSI-driven value:

$$k(m,n) = k_{\min} + \left(k_{\max} - k_{\min}\right) \cdot \mathrm{PSSI}(m,n), \tag{21}$$

with $k_{\min} = 2.5$ (calm sea) and $k_{\max} = 6.0$ (very rough sea) as default values. The detection threshold then becomes:

$$T(m,n) = \hat{\mu}(m,n) + k(m,n) \cdot \hat{\sigma}(m,n). \tag{22}$$

In high-PSSI regions, the raised threshold suppresses the elevated clutter, maintaining constant false alarm rate. In low-PSSI regions, the reduced k maintains sensitivity to small, low-contrast targets that a conservatively high fixed k would miss.

C. Adaptive Elliptical Kernel Geometry

This is the principal novelty of PGA-CFAR. Rather than using a square or annular background window, we define an **elliptical window whose orientation and aspect ratio adapt to the local wave geometry**.

From $\theta_\nabla(m,n)$ and $\|\nabla\mathrm{PSSI}\|(m,n)$, we define an *anisotropy factor*:

$$\rho(m,n) = 1 + \beta \cdot \|\nabla\mathrm{PSSI}(m,n)\|, \tag{23}$$

where $\beta = 3.0$ (default) controls how aggressively the kernel deforms. The background estimation ellipse at pixel (m,n) has:

- **Semi-major axis** $a(m,n) = r_0 \cdot \rho(m,n)$, oriented *perpendicular to* θ_∇, i.e., *along* the wave crests, where clutter statistics are approximately stationary.
- **Semi-minor axis** $b(m,n) = r_0/\rho(m,n)$, oriented *along* θ_∇, i.e., in the wave propagation direction, where roughness varies — so the window is compressed to avoid including heterogeneous cells.

The product $a \cdot b = r_0^2$ is constant, meaning the total number of background cells does not grow unboundedly with ρ. A pixel (m',n') with offset $(x',y') = (m'-m, n'-n)$ is included in the background region if:

$$\left(\frac{x'\cos\theta_\nabla + y'\sin\theta_\nabla}{a(m,n)}\right)^2 + \left(\frac{-x'\sin\theta_\nabla + y'\cos\theta_\nabla}{b(m,n)}\right)^2 \le 1, \tag{24}$$

excluding the guard region (inner ellipse with semi-axes $a_g = a/2$, $b_g = b/2$):

$$\mathcal{R}_{\text{bg}}(m,n) = \mathcal{E}(a,b,\theta_\nabla) \setminus \mathcal{E}(a_g,b_g,\theta_\nabla). \tag{25}$$

The guard region prevents the target itself from contaminating the clutter estimate. When the sea is isotropic (calm, $\|\nabla\mathrm{PSSI}\| \approx 0$), $\rho \to 1$ and the ellipse degenerates to a circle, recovering conventional isotropic CFAR behavior as a special case.

D. Complete PGA-CFAR Decision Rule

Combining both adaptations, the detection decision at each pixel is:

$$\hat{y}(m,n) = \begin{cases} 1 & \text{if } z(m,n) > \hat{\mu}_{\mathcal{R}}(m,n) + k(m,n)\cdot\hat{\sigma}_{\mathcal{R}}(m,n) \\ 0 & \text{otherwise,} \end{cases} \tag{26}$$

where $z(m,n)$ is the detection statistic (Span power), and $\hat{\mu}_{\mathcal{R}}$, $\hat{\sigma}_{\mathcal{R}}$ are the mean and standard deviation estimated over $\mathcal{R}_{\text{bg}}(m,n)$.

A polarimetric pre-screening step further reduces false alarms by masking pixels that cannot physically be ships: only pixels satisfying $\alpha(m,n) > \alpha_{\text{thresh}}$ *and* $H(m,n) < H_{\text{thresh}}$ are admitted as candidates. The complete algorithm is summarized in Algorithm 1.

Algorithm 1 PGA-CFAR Ship Detection Algorithm

Require: Coherency matrix field $\mathbf{T}(m,n)$; incidence angle map $\theta_i(m,n)$
Ensure: Binary detection map $\hat{y}(m,n)$
Compute H, A, α via eigendecomposition of $\mathbf{T}$ *(Sec. III-D)*
Compute ϕ_{HH-VV} via (10); compute $\mathcal{B}_{\text{meas}} = \sigma^0_{HH}/\sigma^0_{VV}$
Compute Φ via (15); compute $\Delta\mathcal{B}$ via (16)
Compute $\mathrm{PSSI}(m,n)$ via (14)
Apply 11×11 Gaussian smoothing to PSSI map ($\sigma_s = 2$ px)
Apply Sobel operator $\Rightarrow \theta_\nabla(m,n)$, $\|\nabla\mathrm{PSSI}(m,n)\|$
Compute $k(m,n)$ via (21)
Compute $\rho(m,n)$ via (23); compute ellipse axes (a,b)
Apply polarimetric pre-screen: candidate mask $= \{\alpha > \alpha_{\text{thresh}}\} \cap \{H < H_{\text{thresh}}\}$
for each candidate pixel (m,n) **do**
  Define $\mathcal{R}_{\text{bg}}(m,n)$ via (24)–(25)
  $\hat{\mu} \leftarrow \mathrm{mean}_{(m',n')\in\mathcal{R}_{\text{bg}}} z(m',n')$; $\hat{\sigma} \leftarrow \mathrm{std}_{(m',n')\in\mathcal{R}_{\text{bg}}} z(m',n')$
  Apply decision rule (26) $\Rightarrow \hat{y}(m,n)$
end for
Morphological opening (3×3) → connected component labelling → size filter → centroid extraction
return $\hat{y}(m,n)$

VI. Complete Methodology

Figure 1 shows the complete pipeline. It consists of five sequential stages described below.

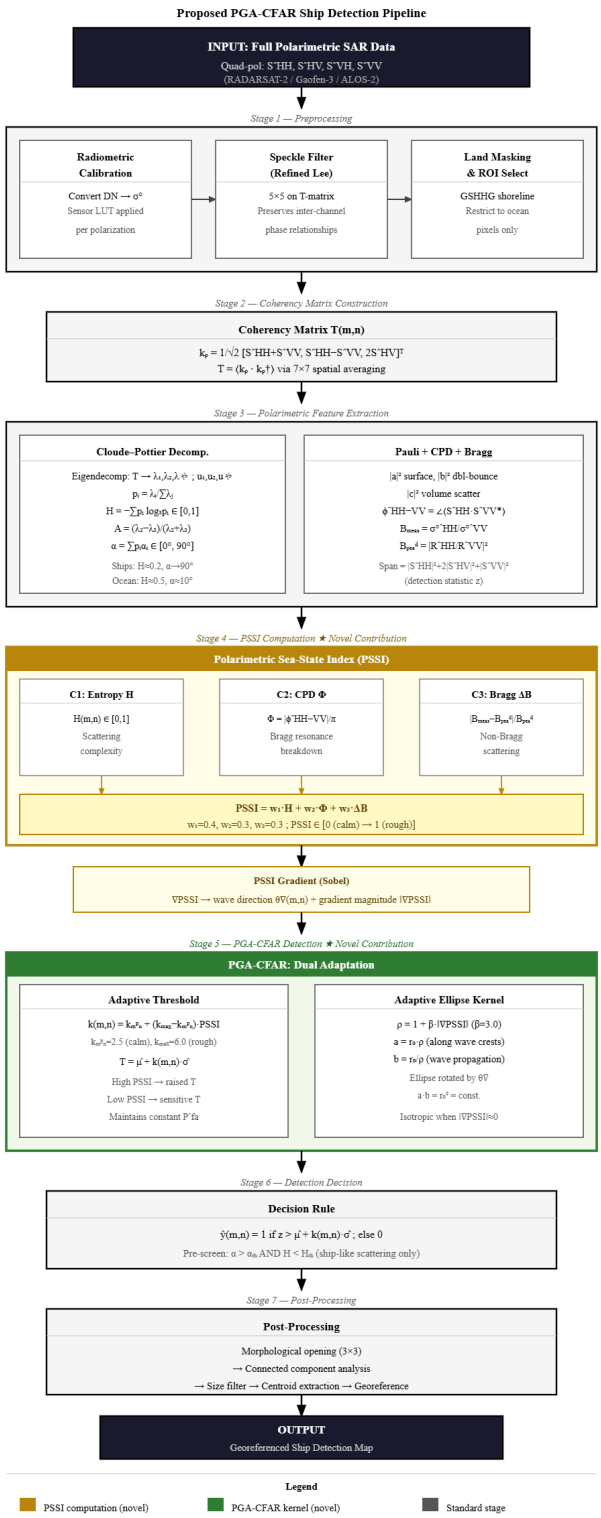

Fig. 1. Block diagram of the proposed PGA-CFAR framework. Stages 4 and 5 (gold and green) represent the novel contributions of this work. The PSSI block feeds both the adaptive threshold path and the adaptive kernel geometry path, coupling all downstream decisions to the local physical sea state.

A. Stage 1: Preprocessing

1) **Radiometric calibration**: Convert raw digital numbers to normalized backscatter coefficients σ^0 using the sensor calibration look-up table. This step ensures that comparisons between different acquisition dates, incidence angles, and sensors are physically meaningful.
2) **Polarimetric-preserving speckle filtering**: Apply the refined Lee filter [21] with a 5×5 window, operating directly on the elements of the 3×3 coherency matrix $\mathbf{T}$ rather than on individual intensity channels. Filtering channels independently would destroy the inter-channel phase relationships that the subsequent polarimetric analysis depends on.
3) **Land masking**: Apply an ocean mask derived from the Global Self-consistent Hierarchical High-resolution Geography (GSHHG) shoreline database, or from a simple normalized difference water index computed from an auxiliary optical band if available, to restrict all processing to ocean pixels.

B. Stage 2: Coherency Matrix Construction

For each pixel (m, n), form the Pauli vector $\mathbf{k}_P$ from the calibrated scattering matrix elements and estimate $\mathbf{T}(m, n) = \langle \mathbf{k}_P \mathbf{k}_P^\dagger \rangle$ via 7×7 spatial averaging. This window size represents a compromise between statistical stability of the estimate and spatial resolution preservation for small ship detection.

C. Stage 3: Polarimetric Feature Extraction

From $\mathbf{T}(m, n)$ and the calibrated channel data, compute at each pixel:

- Cloude–Pottier parameters: H, A, α via eigendecomposition of $\mathbf{T}$ (6)–(9)
- Pauli decomposition powers: $|a|^2$, $|b|^2$, $|c|^2$ (5)
- Co-polarization phase difference: ϕ_{HH-VV} (10)
- Measured Bragg ratio: $\mathcal{B}_{\text{meas}} = \sigma^0_{HH}/\sigma^0_{VV}$
- Span power: $\text{Span} = |S_{HH}|^2 + 2|S_{HV}|^2 + |S_{VV}|^2$

D. Stage 4: PSSI Computation

Using the features above, compute per-pixel PSSI via (14)–(16). Apply 11×11 Gaussian smoothing ($\sigma_s = 2\,\text{px}$) to reduce speckle in the PSSI map while preserving genuine sea-state boundary structure. Then apply Sobel operators to obtain $\theta_\nabla(m, n)$ and $\|\nabla \text{PSSI}(m, n)\|$ across the full scene.

E. Stage 5: PGA-CFAR Detection and Post-Processing

Run Algorithm 1 with parameters $k_{\min} = 2.5$, $k_{\max} = 6.0$, $r_0 = 15\,\text{px}$, $\beta = 3.0$, $\alpha_{\text{thresh}} = 40°$, $H_{\text{thresh}} = 0.6$. Apply morphological opening, connected component analysis, and size filtering (minimum component area $= 4\,\text{px}^2$) to produce the final binary detection map. Extract centroids of retained components and georeference them using the sensor's range-Doppler geometry model.

VII. Dataset Description

The methodology will be validated using quad-polarimetric SAR datasets acquired over ocean regions with confirmed ship presence. Suitable datasets include:

- **RADARSAT-2 Fine Quad-Pol**: C-band (5.405 GHz), $\sim$8 m resolution, 25 km swath. Widely used benchmark for PolSAR ship detection research [13].
- **Gaofen-3**: C-band, available in quad-pol mode at 8–500 m resolution; the first Chinese civilian satellite with fully polarimetric capability [22].
- **ALOS-2 PALSAR-2**: L-band (1.27 GHz), quad-pol at 3 m resolution. L-band Bragg resonance wavelengths correspond to longer ocean waves, providing complementary roughness sensitivity.

Ground truth ship positions will be obtained from AIS (Automatic Identification System) records temporally matched to the SAR acquisitions. AIS transmits GNSS-derived vessel positions at regular intervals and is required by international maritime law for vessels above 300 gross tonnes, providing accurate point-level detection reference targets.

VIII. Expected Outcomes and Evaluation

A. Performance Metrics

- **Probability of Detection** P_d: fraction of AIS-confirmed ships correctly detected.
- **False Alarm Rate** P_{fa}: fraction of non-ship pixels incorrectly flagged.
- **Figure of Merit** $\mathrm{FoM} = P_d - 10 \cdot P_{fa}$: combined metric penalizing false alarms heavily.
- **ROC curve**: P_d vs. P_{fa} over the full range of $k_{\min}$ values, characterizing the detection–false alarm tradeoff.

B. Ablation Baselines

To isolate the contribution of each novel element, PGA-CFAR will be compared against:

1) **CA-CFAR**: fixed square window, fixed k — baseline.
2) **OS-CFAR** [8]: ordered statistics, fixed square window.
3) **H-CFAR**: k adapted by entropy alone, fixed square window — tests PSSI vs. single feature.
4) **PGA-CFAR (threshold only)**: PSSI-adaptive k, but fixed square window — isolates the geometry adaptation contribution.
5) **Full PGA-CFAR**: PSSI-adaptive k + elliptical kernel — proposed method.

The step from (4) to (5) directly quantifies the detection improvement attributable to the adaptive kernel geometry — the core novel mechanism.

C. Expected Outcomes

Based on the physical reasoning developed in this paper:

- False alarms in rough-sea regions: 30–50% reduction vs. CA-CFAR, from adaptive k.
- Detection of near-sea-surface ships: measurable improvement from elliptical kernel alignment, particularly for ships near wave-active regions.
- Clutter transition zones: substantial improvement from kernel compression in the gradient direction, preventing contamination of background estimates.

IX. Implementation Plan

The complete framework will be implemented in MATLAB using the Image Processing Toolbox, Statistics and Machine Learning Toolbox, and Mapping Toolbox. The six software modules are:

1) **Module 1 — PolSAR Reader**: Ingests binary (.bin) or HDF5-format quad-pol SAR data; assembles $\mathbf{T}(m, n)$ matrices via boxcar spatial averaging.
2) **Module 2 — Feature Engine**: Computes H, A, α, ϕ_{HH-VV}, $\mathcal{B}_{\text{meas}}$, Span using vectorized MATLAB operations.
3) **Module 3 — PSSI Mapper**: Implements (14)–(19); outputs PSSI image, gradient magnitude map, and gradient angle map.
4) **Module 4 — PGA-CFAR Engine**: Implements Algorithm 1 with precomputed ellipse membership lookup tables indexed by discrete (θ_∇, ρ) bins, avoiding per-pixel mask recomputation.
5) **Module 5 — Post-Processor**: Morphological operations (`imopen`), connected components (`bwlabel`), size filtering, centroid extraction, and georeferencing.
6) **Module 6 — Evaluator**: Ingests AIS ground truth, computes P_d, P_{fa}, FoM, generates ROC curves, and produces visualization outputs.

X. Discussion

A. Novelty and Significance

To the best of the authors' knowledge, no prior work has proposed using a multi-feature polarimetric sea-state map to simultaneously adapt both the threshold scaling factor *and* the geometric shape and orientation of the CFAR estimation kernel. The use of the PSSI gradient to orient elliptical estimation windows along wave crests is a conceptually new coupling between physical oceanography and statistical signal detection. It addresses a limitation that has existed in CFAR-based maritime SAR detection since the method was first applied to ocean imagery in the 1990s.

B. Generality and Limitations

PGA-CFAR is sensor-agnostic and requires only quad-pol data and a per-pixel incidence angle map, both of which are standard outputs of any modern PolSAR satellite. The main limitation of the current formulation is the single-gradient wave direction estimate, which may not capture multi-directional swell. Future work will explore structure tensor analysis for multi-directional wave fields and extension to multi-temporal PolSAR stacks for change-based vessel detection.

XI. Conclusion

This paper has presented the design of a physically-motivated, training-data-free framework for ship detection in Polarimetric SAR imagery. The key contribution is the **Polarimetric Sea-State Index (PSSI)**: a per-pixel composite of polarimetric entropy, co-polarization phase difference, and Bragg ratio deviation that maps local ocean roughness continuously across a SAR scene. Driving both the threshold and the window geometry of a CFAR detector with this index yields the **PGA-CFAR** algorithm — a detector that knows, at each pixel, how rough the local sea is and in which direction roughness is changing, and uses both facts to make smarter, more robust detection decisions.

The framework is grounded in electromagnetic scattering physics, computationally tractable, and requires no annotated training data — properties that make it particularly well-suited to the practical constraints of operational maritime surveillance. Phase II will deliver the complete MATLAB implementation and experimental validation on multi-scene quad-pol SAR data.

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